

0.1 Question 1

In the general population, approximately 10% of people are left-handed. You randomly sample (with replacement) 77 people from the general population.

0.1.1 Question 1a (3 pts)

Answer each of the following questions using Markdown in the same cell below.

- **i).** Let X be the random variable that is the number of people in your sample who are left handed. Define how X is distributed using the notation: $X \sim \text{DistributionName}(\text{parameter_values})$.
- **ii).** What is the support of X ?
- **iii).** What is the PMF of X ? (Give your answer as a closed-form function)

Answer all of the questions above in the same LaTeX cell below.

i). $X \sim \text{Bin}(77, 0.1)$

ii). $\{0, 1, \dots, 77\}$

iii).

$$P(X = x) = \binom{77}{x} (0.1)^x (0.9)^{77-x} \quad (1)$$

0.2 Question 1d (2 pts)

To plot a histogram of the distribution of X , use `stats.binom.pmf` to define the pmf array in the code below.

Be sure to also fill in the missing parameters in the title of the plot.

```
In [8]: X = np.array(range(78))

pmf = stats.binom.pmf(X, 77, .1)

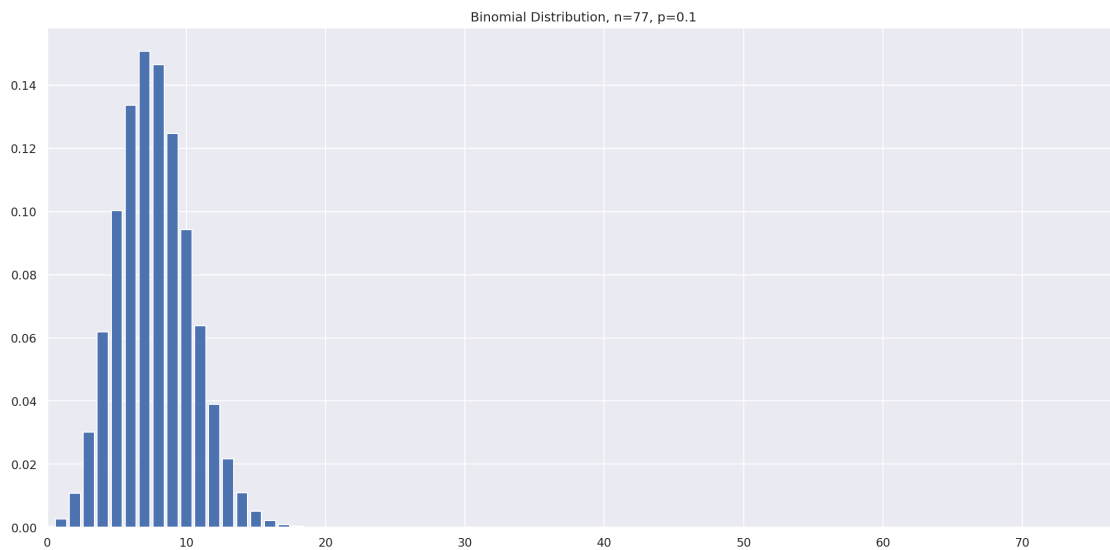
fig, ax = plt.subplots()

# Note: to plot this distribution we can't use .hist, because we don't
# have an array of data,
# Instead we use .bar because we have an array of probabilities

ax.bar(X, pmf);

plt.xlim(0,77)

# Fill in the correct values for n and p in the title:
plt.title("Binomial Distribution, n=77, p=0.1");
```



0.3 Question 1f (5 points)

We can approximate the theoretical distribution of $\text{Bin}(n, p)$ by drawing many random samples from the distribution and plotting a **density** histogram of the sample data.

In the code below, you are going to repeatedly draw random samples from a $\text{Bin}(77, 0.10)$ distribution using sample sizes of 50, 500 and 5000 and plot the resulting empirical distributions of your samples.

You'll then compare this to the theoretical distribution you plotted in 1d.

Fill in all the ellipses (...) in the code below to complete this.

You should see that the empirical distributions approach the theoretical distribution as the sample size increases.

```
In [12]: #Randomly sample 50 values from Bin(77,0.10)
         sample_50 = stats.binom.rvs(n=77, p=0.10, size=50)

         #Randomly sample 500 values from Bin(77,0.10)
         sample_500 = stats.binom.rvs(n=77, p=0.10, size=500)

         #Randomly sample 5000 values from Bin(77,0.10)
         sample_5000 = stats.binom.rvs(n=77, p=0.10, size=5000)

         # Create a figure that will have 4 subplots (4 histograms)
         fig, ax = plt.subplots(nrows=4, ncols=1, figsize = (3,5))

         plt.subplots_adjust(hspace=1)

         # Plot a density histogram of the sample data of size 50
         # ax[0] indicates we are going to plot this distribution in the first out of 4 subplots
         # Use density histogram because we are binning a list of simulated numbers to estimate the pro
         ax[0].hist(sample_50, density=True, bins = np.arange(-0.5, 100, 1))
         ax[0].set_axisbelow(True)
         ax[0].grid(alpha=0.25)
         ax[0].set_xlim([0, 25]);
         ax[0].set_title("Simulated Bin(77, 0.10) distribution using sample of size 50")

         # Use the same format as above to plot
         # a density histogram of the sample data of size 500.
         # Note, to put this as the 2nd plot out of 4, use ax[1] (instead of ax[0])
         ax[1].hist(sample_500, density=True, bins=np.arange(-0.5, 100, 1))
         ax[1].set_axisbelow(True)
         ax[1].grid(alpha=0.25)
         ax[1].set_xlim([0, 25]);
```

```

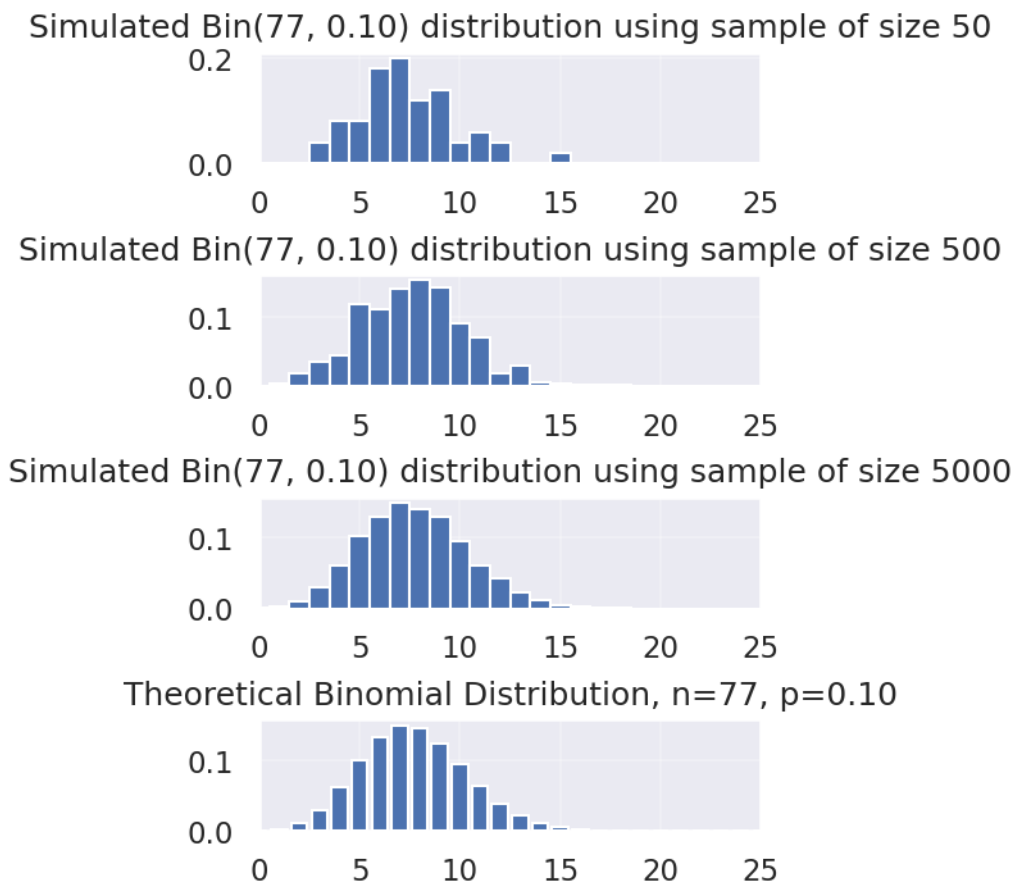
ax[1].set_title("Simulated Bin(77, 0.10) distribution using sample of size 500")

# Plot a density histogram of the sample data of size 5000
ax[2].hist(sample_5000, density=True, bins=np.arange(-0.5, 100, 1))
ax[2].set_axisbelow(True)
ax[2].grid(alpha=0.25)
ax[2].set_xlim([0, 25]);
ax[2].set_title("Simulated Bin(77, 0.10) distribution using sample of size 5000")

# Plot the actual theoretical distribution. Use bar because we know the exact probabilities:
ax[3].bar(X, pmf)
ax[3].set_axisbelow(True)
ax[3].grid(alpha=0.25);
ax[3].set_xlim([0, 25]);
ax[3].set_title("Theoretical Binomial Distribution, n=77, p=0.10")

plt.show()

```



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0.4 Question 3 (9 points)

Let X be a continuous random variable with support $1 \leq x \leq 7$.

The graph of the PDF $f(x)$ for X is shown here:

a). What is $P(3.25 \leq X \leq 5)$? Show all work using LaTeX and give your final answer as a fully simplified fraction.

b). What is $E[X]$? Show all work using LaTeX and give your final answer as a fully simplified fraction.

Answer both questions in the same cell below using LaTeX.

a)

First note that the range $[3.25, 5]$ in f contains a constant function from $[3.25, 4)$ and a downward slope from $(4, 5]$.

Area of the first part:

$$(4 - 3.25) \cdot 0.25 = 0.1875 = \frac{3}{16} \quad (2)$$

Area of the second part can easily be solved via an integral (shifted over to the origin for simplicity):

$$\int_0^1 \left(-x \frac{0.25}{3} + 0.25 \right) dx = \left(-x^2 \frac{0.25}{6} + 0.25x \right) \Big|_0^1 \quad (3)$$

$$= -\frac{0.25}{6} + 0.25 = \frac{5}{24} \quad (4)$$

Final sum:

$$\frac{3}{16} + \frac{5}{24} = \frac{19}{48} \quad (5)$$

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0.5 Question 4 (8 points)

A factory manufactures solid state hard drives for Seagate Technology.

For each scenario described below: - i). State the distribution you could use to model this random variable in the form $RandomVariable \sim DistributionName(parameter_values)$

- ii). State any assumptions you need to make to use this distribution to model this situation.
- iii). State the support of the Random Variable.
- iv). Give the probability distribution for the random variable (i.e. either the pmf or pdf) as a closed-form function.

SCENARIOS:

- a). Approximately 4% of hard drives that come off of a particular assembly line are defective. Let Y be the number that are defective in a batch of 300.
- b). Approximately 4% of hard drives that come off of a particular assembly line are defective. Let W be whether or not a hard drive is defective.
- c). On average, there are 8 defective drives per day. Let X be the time (in hours) between defective drives.
- d). On average, there are 8 defective drives per day. Let M be the number of defective drives in a day.

Answer all parts (a-d) below in the same cell:

- a)
- b) $Y \sim Bin(300, 0.04)$

- ii) That the percentage change of a hard drive being defective is the same over the course of the full 300.
- iii) Support is $[0, 1, 2, \dots, 300]$
- iv) $P(Y = k) = \binom{300}{k} \cdot 0.04^k \cdot (0.96)^{300-k}$

- b)
- c) $W \sim Bern(0.04)$

- ii) One drive being defective does not affect others.
- iii) 0 (functional) and 1 (defective).
- iv) $P(W = k) = 0.04^k \cdot 0.96^{1-k}$
- v)
- vi) $X \sim \text{Exp}(\frac{8}{24})$
- vii) Events are independent of each other.
- viii) $\mathbb{R}^+$
- ix) $P(X = k) = \frac{1}{3}e^{-\frac{k}{3}}$
- x)
- xi) $M \sim \text{Poi}(\frac{8}{24})$
- xii) Events are independent of each other.
- xiii) $\{0, 1, \dots\}$
- xiv) $P(M = k) = \frac{\frac{1}{3}^k e^{-\frac{1}{3}}}{k!}$

0.5.1 Question 5a (4 points)

What is the probability that a flood will occur within the next 90 years? Fully justify/show all steps and write your answer using LaTeX. Round your final answer to 3 decimal places.

Let $X \sim \text{Exp}(\frac{1}{500})$ represent the number of years before a flood occurs. We need to find $P(X < 90)$

$$P(X < 90) = F(90) = 1 - e^{-\frac{90}{500}} \approx 0.165 \quad (6)$$

Thus the probability that a flood occurs in next 90 years is about 16.5%.

0.6 Question 6a (2 points)

Graph the pdf of W on the domain $400 \leq W \leq 600$

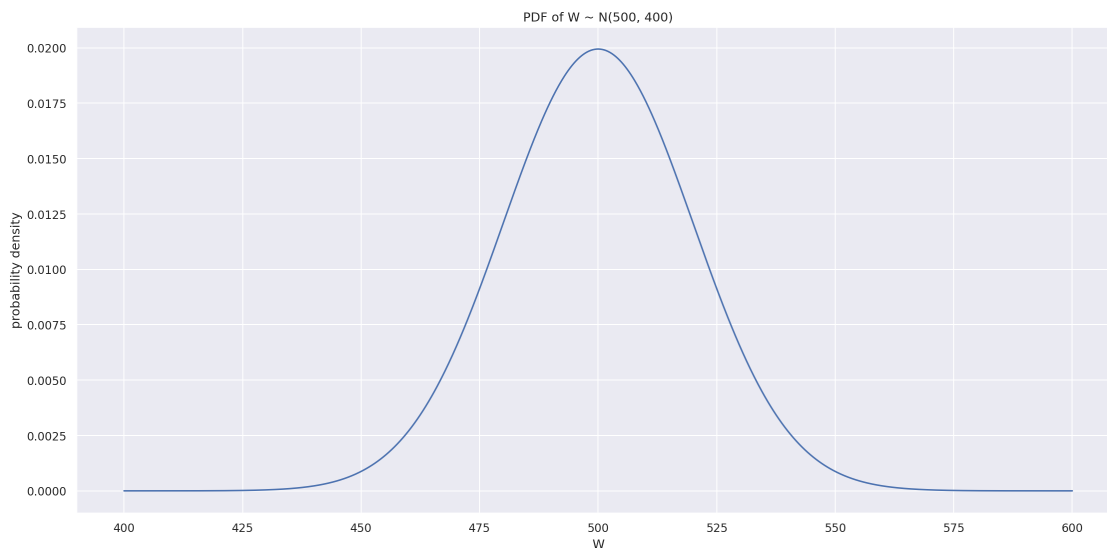
(Hint: use `plt.plot` and an appropriate method from `scipy.stats.norm`: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.norm.html>). Be sure to carefully read what parameters the method takes as inputs.

```
In [41]: # Create a vector of w values between 400 and 600
w = np.linspace(400, 600, 1000)
dist = stats.norm.pdf(w, 500, 20)

plt.plot(w, dist)
# Your code above this line

plt.title("PDF of W ~ N(500, 400)")
plt.ylabel("probability density")
plt.xlabel("W")

plt.show()
```



0.7 Question 6c (3 points)

i). What is the probability the actual costs are between \$475 and \$515? Use an appropriate method from `scipy.stats.norm` to calculate this: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.norm.html>

ii). Then show graphically what this number means in terms of the PDF (for full credit your output should have the same format and labels as the plot shown directly above).

```
In [48]: q6ci_answer = stats.norm.cdf(515, 500, 20) - stats.norm.cdf(475, 500, 20)
         q6ci_answer
```

```
Out[48]: np.float64(0.6677228739562764)
```

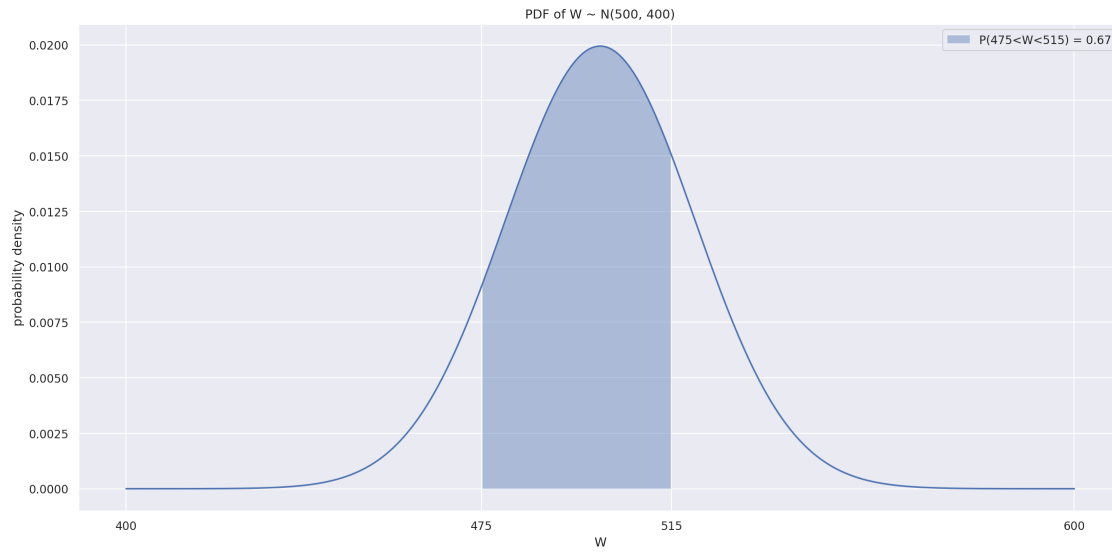
```
In [51]: # Code to create graph for 5c(ii)
```

```
plt.plot(w, stats.norm.pdf(w, 500, 20))

(
    plt.fill_between(w, stats.norm.pdf(w, 500, 20),
                     where=[(w > 475 and w < 515) for w in w],
                     label='P(475<W<515) = '+str(round(float(q6ci_answer), 2)),
                     alpha=0.4)
)

labels = [400, 475, 515, 600]
plt.xticks(labels)
plt.legend()
plt.title("PDF of W ~ N(500, 400)")
plt.ylabel("probability density")
plt.xlabel("W")

plt.show()
```



```
In [49]: grader.check("q6c")
```

```
Out[49]: q6c results: All test cases passed!
```